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Breast Cancer Detection with Machine Learning

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Abstract: Breast cancer is one of the leading cause for the death of women. In women Breast cancer is treated as the most significant issue. According to statistics released by the International Agency for Research on Cancer (IARC) in December 2020, Breast cancer has now overtaken lung cancer as the most commonly diagnosed cancer in women worldwide. Early diagnosis of this helps to prevent the cancer. If breast cancer is detected in early stage, then Survival rate is very high. Machine Learning methods are effective ways to classify data. Especially in the medical field, where those methods are widely used in diagnosis and analysis for decision making. In this paper, Data Visualization and performance comparisons between different machine learning algorithms: Support Vector Machine (SVM), Decision Tree, Naive Bayes (NB), K Nearest Neighbours (k-NN), Adaboost, XGboost and Random Forest conducted on Wisconsin breast cancer Dataset. The main objective is to evaluate the accuracy in the classification of data in terms of efficiency and effectiveness of each algorithm in terms of accuracy, precision, sensitivity and specificity. Our aim is to review various Techniques To detect early, efficiently and accurately Using Machine Learning. Experimental results show that XGboost offers the highest accuracy (98.24%) with the lowest error rate.

Keywords: Breast Cancer, Machine Learning, Wisconsin, Algorithms, Detection

I. INTRODUCTION

Total number of women dying in 2021 is approximately 963,000, according to the World Health Organization (WHO). Still, the organization predicts that the number could reach 2.9 million globally. Breast cancer can occur in women and rarely in men. The ICMR (Indian Council of Medical Research) recently published a report which stated that in 2020 the total number of new cancer cases is expected to be about 17.3 lakhs. An Indian woman is diagnosed with breast cancer in every four minutes. Breast cancer is a disease that occurs but when a woman or a man is aware of this symptom, it immediately goes beyond its original stage. Breast cancer is a common and dangerous disease in women, cancer is the creation of abnormal cells that come into these cells genetically and mutated. Spreads throughout the body, leading to death in diagnosis and treatment. There are two types of breast cancer, Malignant and Benign. The first is classified as harmful has the ability to infect other organs and is cancerous, Benign is classified as non-cancerous. This disease infects the women's chest and specifically glands and milk ducts, the spread of breast cancer to other organs is frequent and could be through the bloodstream. Different techniques are used to capture breast cancer such as Ultrasound Sonography, Computerized Thermography, Biopsy (Histological images).

Machine learning and Data mining techniques are straightforward and effective ways to understand and predict data. Radiologist examines and analyses himself and then he / she decides the result after participating with other experts. This process takes time and the results depend on the knowledge and experience of the staff. In addition, experts are not available in every field of the world. Therefore, the research community proposed automatic A system called CAD (Computer-Aided Diagnosis) for better classification of tumours, accurate results and faster Implementation without the need for radiologists or specialists. Machine learning algorithms (MLs) are indicated as one Option of human vision and experience to make final decisions with high accuracy.

Cancer in women always has a huge incidence rate and mortality rate. Breast cancer alone is estimated to account for 25% of all new cancer diagnoses worldwide and 15% of cancer deaths in women worldwide, according to the latest cancer statistics. Every 1 in 8 Women in USA develop breast cancer in her lifetime. In case of any sign or symptom, people usually visit a doctor immediately, who may refer you to an oncologist for help. An oncologist can diagnose breast cancer by: Examining the patient's medical history thoroughly, examining both breasts, and even checking for swelling or hardening of any lymph nodes in the armpits. Here in this project, we have used the Wisconsin Breast Cancer Dataset (WBCD) and with the dataset we have used machine learning algorithms to predict whether a patient has breast cancer. This paper compares the classification algorithms using an assembled approach suitable for demonstration and direct interpretation of their results. We are using the XGboost classified approach to compare the other classification algorithms and have analysed the accuracy of each classification of the best fit for breast cancer prediction.

II. LITERATURE SURVEY

- A. Turgut Machine learning procedure compared with SVM, KNN, DT, Logistic Regression, Random Forest, ADA Boost. In this various method checked and conclude that highest efficiency is 89% of random forest.
- B. Narasingarao.M presents a survey of the work conducted to detect breast cancer using with different algorithm and conclude the efficiency of algorithm.
- C. Junaid Ahmed achieved 84.21% accuracy by using Adaptive Reasoning Theory, the Wisconsin data set was used, that contains 569 rows of data, and also contains 32 attributes.
- D. Nithya [13] applied the three categorizing methods such as Decision Tree, k-Nearest Neighbour, and Naïve Bayes for the different datasets. The authors also inspect the evaluation metrics of error rate. The implementation was focused on a type of attribute of a dataset.
- E. Shilpa M and C. Nandini [19] implemented the algorithm using python and tested the same using dataset and achieved an accuracy of 94.74 and also reduces the time taken.
- F. Hafizah [2] compared SVM and ANN using four different datasets of breast cancer. The researchers have demonstrated that SVM was better than ANN in performance and result.
- G. S. Gc [1] worked on extracting features including variance, range, and compactness. They used SVM classification to analyse the performance. Their findings showed the highest variance of 95% and compactness 86%. According to their results, SVM can be considered as an appropriate method for Breast Cancer Prediction.

III. METHODOLOGY

A. Dataset Description

We have obtained Breast Cancer Wisconsin (Diagnostic) Dataset from Kaggle. Here 569 Patient's Data Was used for analysis, each instances have 32 Attributes with Diagnosis and Features. Each instance has a parameter of the cancerous non-cancerous cells and we will predict the cancer just by the input of features.

The values of features is in Numeric Format. The 'Target' means the patient Who is having Whether 'Benign' or 'Malignant' Cancer state. Benign means the patient is not having Cancer and Malignant means the patient is having Cancer.

TABLE I
Type of Patients

Patient Type	Target
Benign	1
Malignant	0

B. Data Visualization

We are going to Visualize our Numeric data with Respect to Two categories 1) Benign 0) Malignant.

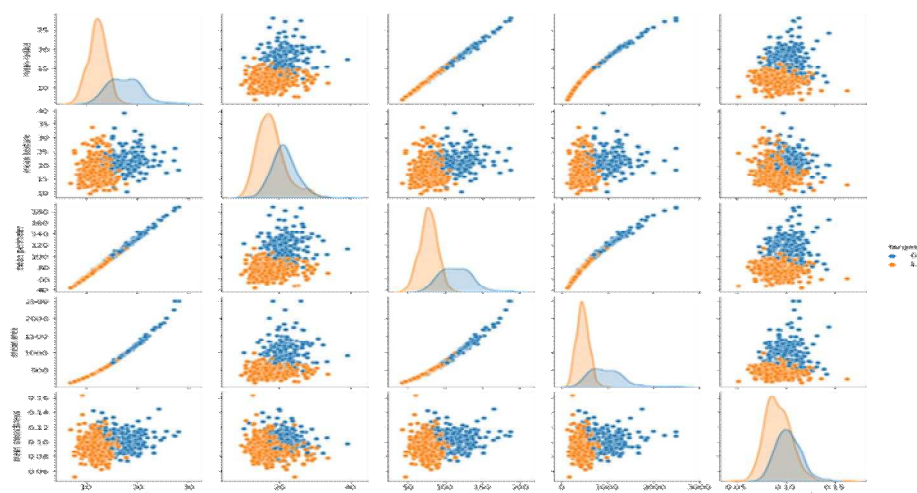


Fig. 1 Pairplot Of Features[mean_radius,mean_texture,mean_perimeter,mean_area,mean_smoothness]

C. Section Headings

We used [Google Colab](#) and Microsoft Visual Code as a Coding platform and get a prediction output from the [Flask](#) in Local Server. Our Methods Includes Supervised Learning Algorithms and Classification Techniques like Support Vector Classifier (SVM), Random Forest, Naïve Bayes, Decision Tree, KNN, Adaboost and XGboost. Dataset contains features which highly vary in units and magnitudes. So, it is required to bring all features to the same level of magnitudes. We did that by using [Standard Scaling](#) in [SKLearn](#).

Model selection is the most important step in Machine Learning. Machine Learning algorithms can be classified as: supervised learning and unsupervised learning. For Our project, we only need supervised learning. We used all Methodologies to Predict the result and Noted their Accuracy.

TABLE II
Comparison Between Techniques

Techniques	Accuracy Without Standard scale	Accuracy With Standard scale
SVM	57.89%	96.49%
KNN	93.85%	57.89%
Random Forest	97.36%	75.43%
Decision Tree	94.73%	75.43%
Naïve Bayes	94.73%	93.85%
Adaboost	94.73%	94.73%
XGboost	98.24%	98.24%

D. Confusion Matrix and Accuracy

Confusion Matrix is used for evaluating the performance of a classification model. The Matrix compares the actual target values with predicted values by the machine learning model. It shows the ways in which your classification model gets confused when it makes predictions.

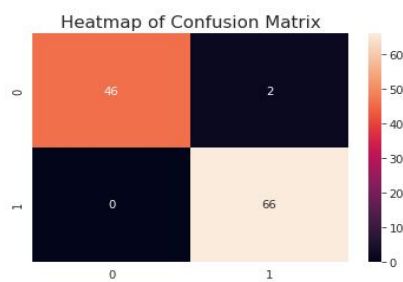


Fig. 2 Confusion Matrix

Accuracy Is given by:

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)}$$

$$= \frac{(46+66)}{(46+66+0+2)} \times 100 = 98.24$$

Where TP= True Positive , TN= True Negative, FP= False Positive, FN=False Negative

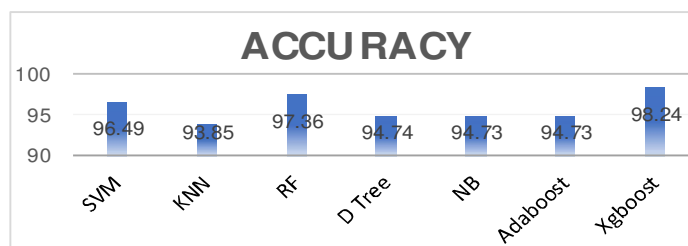


Fig. 3 Accuracy Comparison

IV. PROPOSED SYSTEM ARCHITECTURE

As Shown in diagram, we first Uploaded dataset From Wisconsin Breast Cancer Dataset. After that We did Preprocessing to the data and applied Machine Learning Models, which is used in this project to predict Breast cancer.

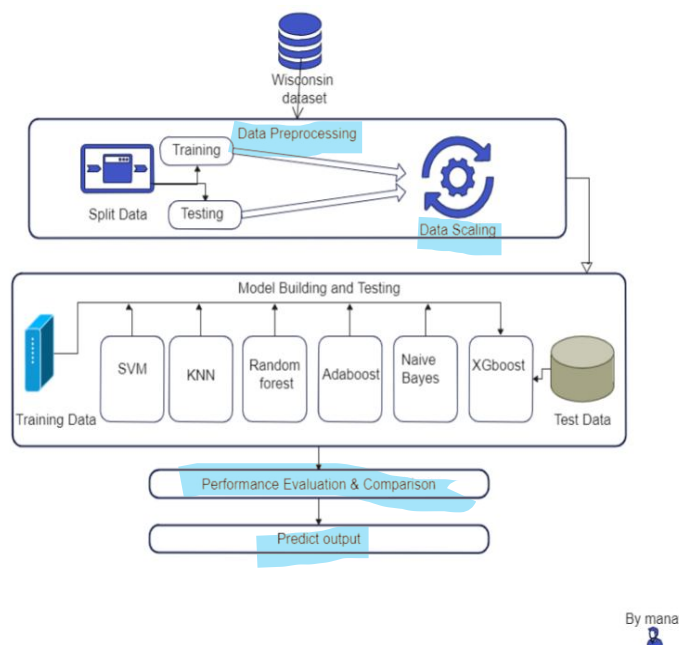


Fig.4 System architecture

V. CONCLUSION & FUTURE WORK

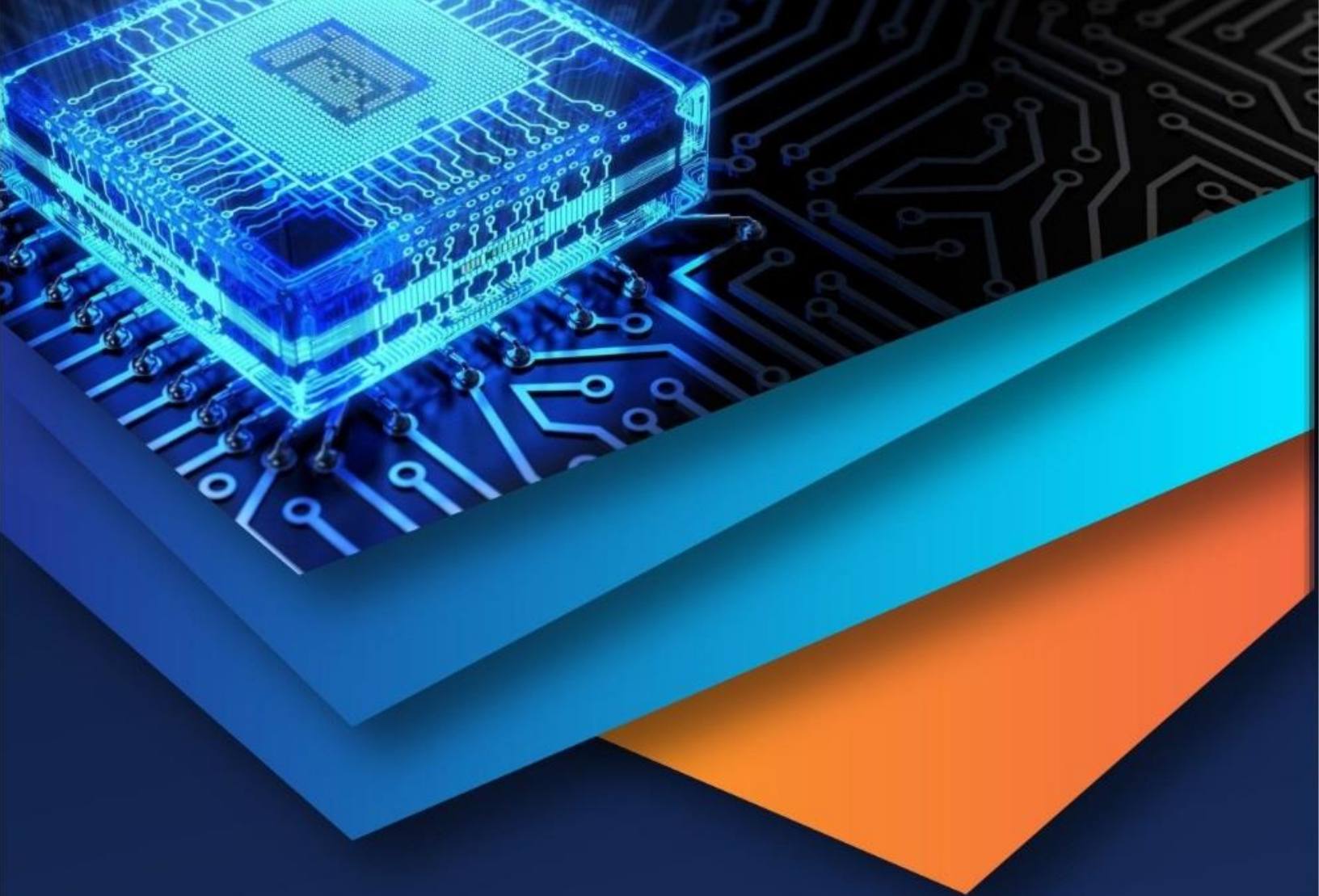
This paper examined different machine learning techniques for detection of breast cancer. The objectives of our study were to analyse the Wisconsin breast cancer dataset by visualizing and evaluating Machine Learning Predictions. With this research paper we can see that among Naïve Bayes, Support Vector Machine, Adaboost, Random Forest Classifier, KNN, Decision Tree, XGboost etc. We concluded that XGboost is the most accurate algorithm for best accurate result for detection of breast cancer with the efficiency of 98.24%. However, it is required that before running the algorithm, the dataset must be pre-processed. In future, we like to add larger dataset and check the efficiency and scalability of algorithm.

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